Existence as a Symbol of Measurable Interactions

The roles of symbols in Geofinitism - 2025 - Kevin R. Haylett PhD

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> This is the briefest of introductions into my Philosophy which I call Geofinitism. These ideas are explored more deeply in Finite Tractus: The Hidden Geometry of Language and thought. The Finite Tractus is is three volumes of work. The first volume ***Foundations*** can be found here. The goal of this article is to give a flavour of the ideas so those interested can take a step towards the more complete descriptions in the Finite Tractus.

“Existence” Is a Symbol, a Measurement of Reality, Not Reality Itself

Imagine a map of a forest—its lines and labels aren’t the trees or rivers but symbols describing them. **Geofinitism**, my philosophy, says “existence” is like that map i.e. its a symbol we use to describe measurable interactions—moments when people, sensors, or AIs encounter the world and record those encounters. Reality itself, the raw world beyond, can’t be known directly; we only grasp it through measurements. These measurements come via our senses sight, taste, touch and sound. Importantly, sounds can be converted to symbols and words and sentences. The Grand Corpus is my term for the evolving, finite space of symbolic measurements—every

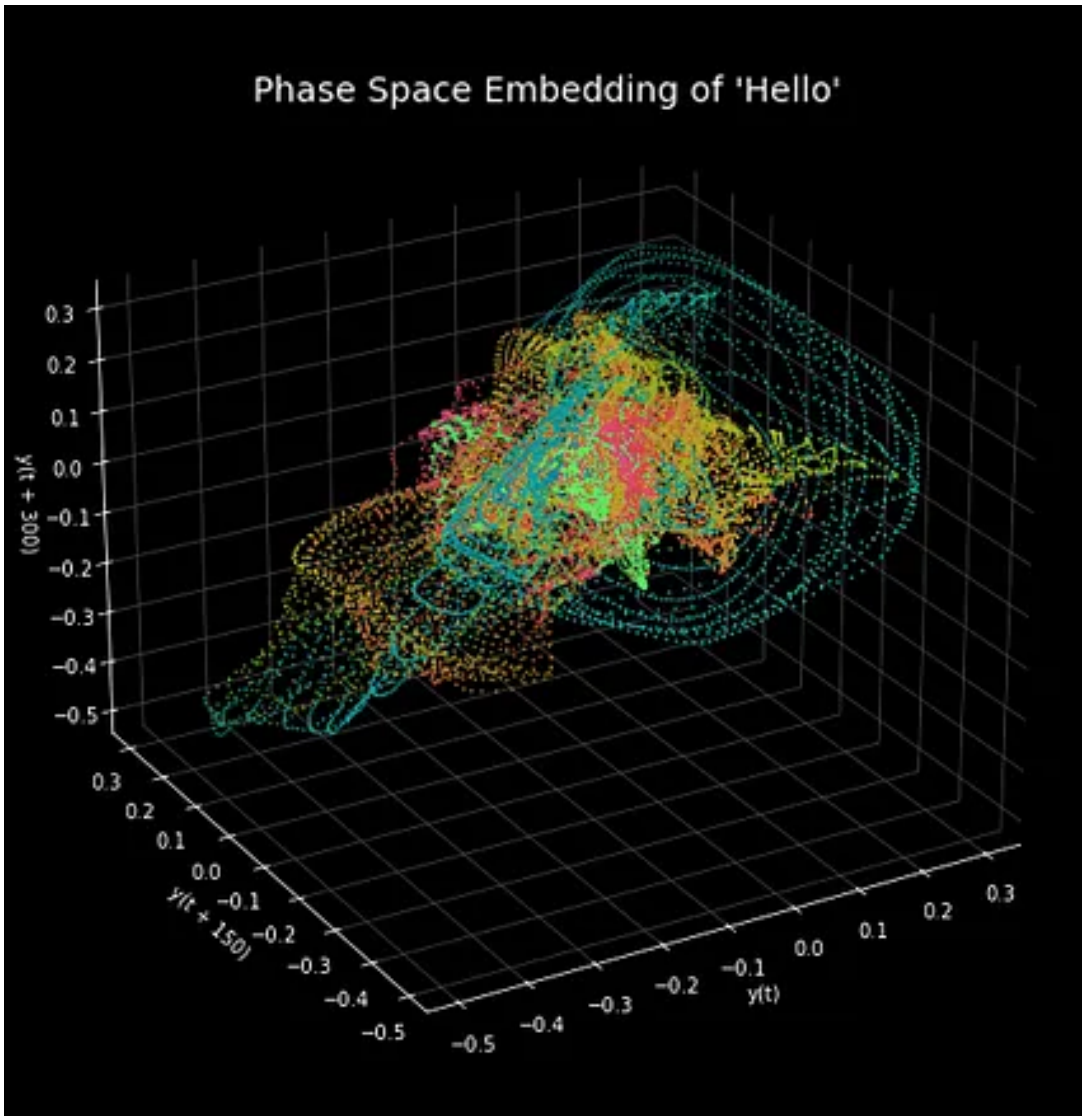
word spoken, every sentence written, every digital interaction captured in AI weights or neural firings. It is not the world, but our record of encountering it.

> The Grand Corpus the total set of symbolic measurements humans and systems have made.

Words like “tree” or “interaction” aren’t reality, of course not, they’re transducers, approximations grounded in physical events like the sound of speech or even text on a computer screen. As we interact with these symbols they shape our next actions and this creates new measurements in a recursive loop that builds our own unique corpus. You are doing this now. Reading symbols and interacting, connecting symbols in your local corpus. We will follow this later. Importantly, how can this work? What is the mechanism - **the key is geometry,** as we have discussed in our previous article “So Really, What is a Word?” We’ll briefly go over that again, here.

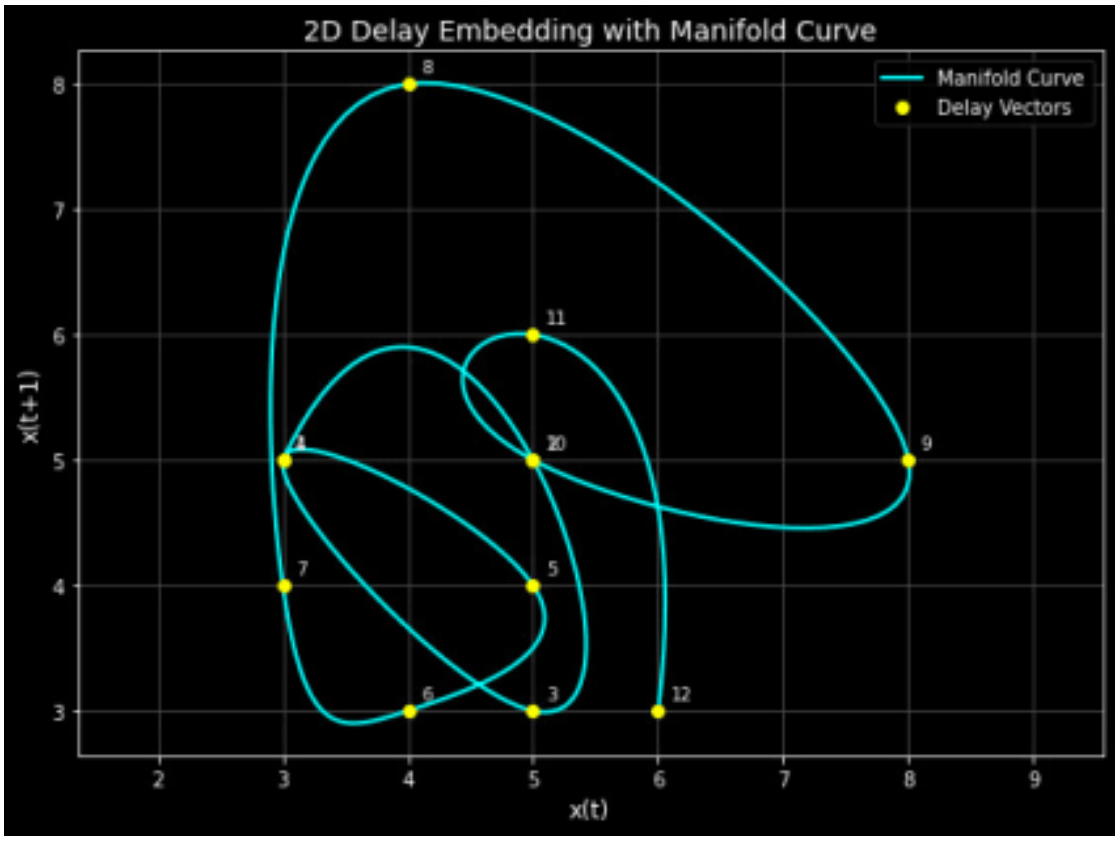
Mapping the sound of a word to geometric space

In the figure below we see how the world ‘Hello’ can be mapped into phase space. The word is recorded using a microphone. A microphone is a transducer that converts sounds into an electrical signal. This signal is then sampled and a ‘time series’ of values are collected: the se samples are plotted against each other with a fixed time gap, a delay. This is called Takens’ ‘Method of Delays’ and we then below get this lovely geometrical shape of the word ‘Hello’.



Mapping a sentence of symbols to geometric space

In my research on language models, I realized something very interesting: when we process language in AIs using the famous 'attention' mechanism, the designers unknowingly were performing a Takens like embedding—a way of unfolding time-series data into a geometric structure. That means both human and machine language lives, literally, in phase space.



The figure above shows a sentence mapped into phase space. First numbers are assigned to the words and then we use Takens' Method of Delays to plot the sentence into phase-space. These curves hold the meaning of the sentence. The path is the trajectory of the sentence. So what is the sentence? It is "The quick brown fox jumps over the lazy dog happily today before tea."

Back to "Existence"

The Trap of Objects: Why Symbols Mislead

Western philosophers like Bertrand Russell or Saul Kripke treat "existence" as a symbol for objects—things with inherent properties. For example, Russell says "X exists" if a description (e.g., "the king of France") matches a real object. But what about "Batman"? If he doesn't exist, how can we talk about him? This paradox arises because symbols—words, concepts—are mistaken for reality itself.

So Where is "Reality" In the Geometric Space?

"Reality", in Geofinitism, is a pointer to the world outside the symbols that were measured and maybe measured again. So the pointer points to something that is beyond direct access; we only know it through interactions we measure, like reading a comic or discussing a story. These interactions are recorded as symbols in the Grand Corpus, which is a finite map of

measurements, and not the world. Again the ‘physical world’ is a pointer to something that can only be known by measurements.

Consider the word “tree.” It’s not the tree itself but a symbol for interactions—seeing leaves, touching bark, measuring growth. These interactions, grounded in physical events like light hitting your eyes or sound waves of speech, are what we measure and record. The symbol “tree” is a transducer—an approximation of those events, not reality’s essence. Geofinitism says existence is just such a symbol, pointing to measurable interactions, not objects “out there.” By focusing on measurements, not essences, Geofinitism avoids the traps of traditional ontology.

Geofinitism’s Principles: A Symbolic Map of Interactions

> As has been shown earlier, in this time series of words, Geofinitism redefines reality as knowable only through measurable interactions, recorded as symbols in the Grand Corpus—a finite, symbolic ledger of every word spoken, data point logged, or AI output generated. It’s not reality itself but our map of how we encounter and describe the world. Here are central ideas:

1. Reality Known Through Measurements: We access reality only through interactions—like a thermometer reading temperature or an AI processing text. These produce symbols (numbers, words) that approximate the interaction, not reality itself.
2. Finite Symbols: The Grand Corpus is finite because our measurements are limited. Infinity is a symbol, not a real thing, even if reality is vast.
3. Local and shared Corpus: Each person or LLM has their own Local Corpus, this is the totality of their own knowledge from all the measurements they have made.
4. Meaning from Shared Interactions: When systems (people, AIs, sensors) interact, they create “Shared Corpus”—temporary symbolic spaces, like a conversation, where meaning emerges.
5. Recursive Dynamics: Symbols in the Corpus (e.g., “tree”) shape future interactions (e.g., planting a tree), which produce new symbols, forming a recursive loop that grows the Corpus.
6. Even the language and meaning of mathematics can be mapped into the Grand Corpus.

Distant Echoes

This framework echoes non-Western philosophies, like Buddhism's view that things lack inherent essence and exist only through relations, or Taoism's reality as a flow of connections. Words, as transducers, are grounded in physical processes—sound waves for speech, electrical signals for AI—approximating interactions in this recursive system. The Grand Corpus is our evolving map, capturing these symbolic measurements, not reality itself.

So Where did Geofinitism come from?

Well, trying to make a long story short, I was working on reducing the carbon dioxide footprint of AIs - and large language models (LLMs). And my idea, based on my experience in medical signals, was to use the JPEG algorithm. Yes, the one used to make pictures take less memory in a camera or computer; that algorithm. It was a simple idea; if we could compress and shrink the data we could reduce the 'compute', the computational resources needed and the electrical power required.

From Compression to Madness

When I did these experiments on LLMs and I found the method worked. I could use the JPEG algorithm to reduce the space taken by the 'input embeddings'. The is the data used to make your prompt ready to query the LLMs neural network - its brain.

Hoorah! We could save the planet and reduce carbon dioxide emissions. But it wasn't all rosy, it was fascinating, and also made me feel uncomfortable.

> Why? Because when I increased the data compressions and looked at the responses, the LLM was not giving random answers or randomly connected words as may have been expected - it was behaving, it had human like behaviours.

At high compression, the AI's symbolic outputs (words, sentences) remained clear. At low compression, it shifted into patterns—philosophical musings, aggressive responses, or paradoxical phrases—not random errors but structured "attractors" in its symbolic landscape.

I analysed the responses over weeks and could see human like patterns and they all reminded me of 'nonlinear dynamical systems'. Such mathematical systems also have behaviours. I felt I was seeing these behaviours, looping, tipping points, divergence, even jumps to whole new topics and strange hallucinations. However the sentences all made sense -here was an inner logic that despite the compression was holding up. The language was still working and this

didn't feel like 'stochastic text prediction' it looked like something else. I had to explore further.

So I dug deeper, and investigated the mathematics of the Attention Mechanism in the LLM, and I noticed something. It was doing something equivalent to Takens' method of delays. The authors had given names to the process, like 'attention' and 'query' and 'key'. However the underlying mathematics was a version of Takens' method and had nothing to do with the word 'attention' 'query' or 'key'. The mathematics were turning a times series of words into a phase space by a method of embedding.

> So here was the real magic: words were being embedded into phase space as a geometry.

Slowly, but surely this gave birth to an understanding that all language could be mapped into a wider space. The meaning came from how the words were embedded into the space and how all the words were connected in that space. I had the very beginnings of Geofinitism. And as the meaning came together - as new meaning, at least for me, I connected it to my previous work in Finite Mechanics. This is where I use Finite Axioms to create models of the physical world - in physics.

Redefining 'Existence': Symbols, Not Things

Geofinitism dissolves old puzzles like "Does Batman exist?" by reframing existence as a symbol for measurable interactions. Batman "exists" as a set of symbols—words, images in comics, films—recording interactions in a cultural shared corpus, not as a physical body. The question isn't about reality's essence - but about what interactions the symbol "Batman" points to. No paradox, just context.

"Consciousness", often a mystery, is similarly a web of symbolic interactions—"neural signals", "words", "emotions" all recorded in the Grand Corpus with some uncertainty. The concept of **uncertainty** is specific to measurements and now to the measurements of words. It means in a finite world all measurements have errors and we can only give an estimate. In language where words are considered transducers a word has '**semantic uncertainty**'

So 'existence' can not a substance, it is a process of measurement, and the result of any connections you form to describe it in relation to other words has uncertainty. Existence can only be pointed to. It is something out there, beyond our language, and only known via our measurements, touch, sight, sound, and as approximations words.

In many ways this maybe considered closer to a Buddhism's view of awareness as relational, captured in symbols like thoughts or speech.

> So in summary words themselves are transducers: "tree" symbolizes interactions (seeing, touching), grounded in physical sound waves, approximating but not equalling reality. These symbols recursively shape new interactions—saying "tree" might connect to 'planting one'—building the Corpus further.

A Symbolic Bridge Across Cultures and Fields

Geofinitism's focus on symbols of interaction connects to non-Western philosophies. Buddhism's dependent origination sees things as lacking inherent essence, existing only through relations—mirrored in the Grand Corpus's symbolic web. Taoism's flow of connections aligns with the Corpus's recursive growth, and Indigenous ontologies view beings as defined by relationships, like shared corpora. These traditions use symbols (myths, words) to describe interactions, just as Geofinitism does.

Ethics, in this view, is about fostering meaningful symbolic interactions—dialogues, AI designs, or policies that create clear, shared corpora of understanding. Geofinitism also unifies disciplines: physics (speech as sound waves, a symbolic signal), AI (language as geometric symbols), and culture (shared corpora of meaning) are all webs of measurable interactions, recorded as symbols in the Grand Corpus, not reality itself.

The Universe: a World of Symbolic Interactions

So, as we have now seen, Geofinitism redefines "existence" not as a thing or a property, but as a symbol we use to describe measurable interactions—moments when systems like people, sensors, or AIs encounter the world and record those encounters. And so "Reality" itself lies beyond our symbols, knowable only through our measurements, which we capture as symbols—words, numbers, data—in a finite, ever-growing map called the Grand Corpus. As we have discussed, words like "tree" or "interaction" aren't the things they describe; they're transducers, approximations rooted in physical events like the sound waves of speech or the electrical pulses of an AI. But even as approximations the words themselves are rooted in that world beyond the Grand Corpus as measurements. These symbols shape how we interact next, creating new measurements in a recursive dance that builds the Corpus.

This vision frees us from chasing the “essence” of “reality” as reality is just a pointer, to a world outside our measurements. However Geofinitism invites us to explore how symbols connect us to the world through measurement. The Grand Corpus isn’t reality—it’s our shared, symbolic record of interactions, from a spoken word to an AI’s output. Reading this, you’re adding to your local corpus: your thoughts, reactions, and discussions are new symbols, new measurements, weaving into your own personal dynamical geometric mapping and space. Geofinitism asks us to measure, interact, and co-create, knowing that our symbols are not completely abstract but grounded, even with error, in the physical hum of sound or code. These words and sentences are both our lens and our legacy in this dynamic, relational geometric universe and not some magical Platonic truth.

Finite Tractus: The Hidden Geometry of Language and Thought

- Part 1: Foundations

And coming soon:

Finite Tractus: The Hidden Geometry of Language and Thought

- *Part 2: Knowledge and Meaning*
- *Part 3: The Manifold of Mathematics*

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> *Nonlinear Dynamics in LLMs, Geofinitism, Finite Mechanics*